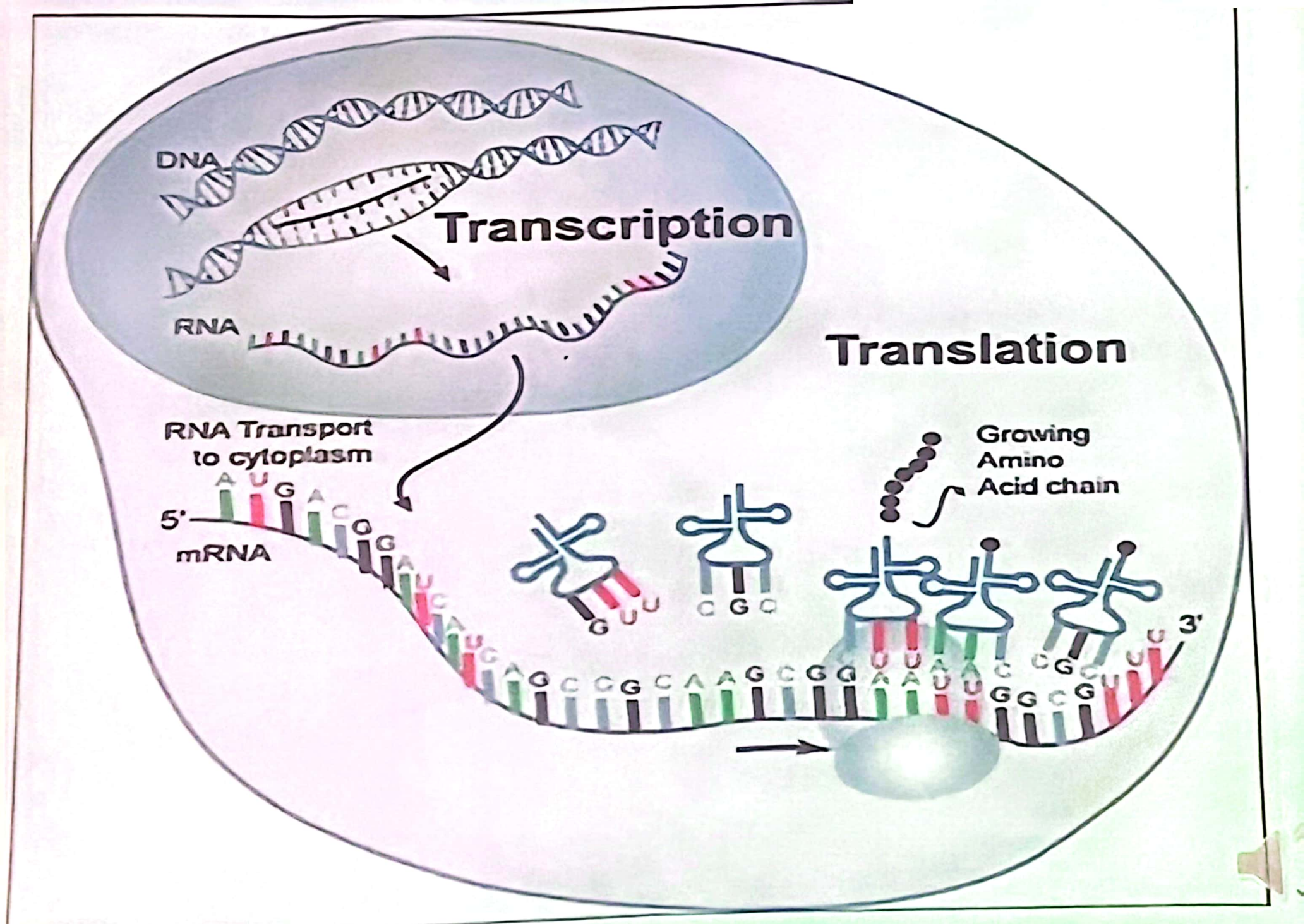


The genetic code





THE GENETIC CODE

Question

How an RNA molecule consisting of only four types of nucleotides

(A, U, C and G)

can specify 20 amino acids?

answer this question, which led to the discovery of the

Genetic Code.

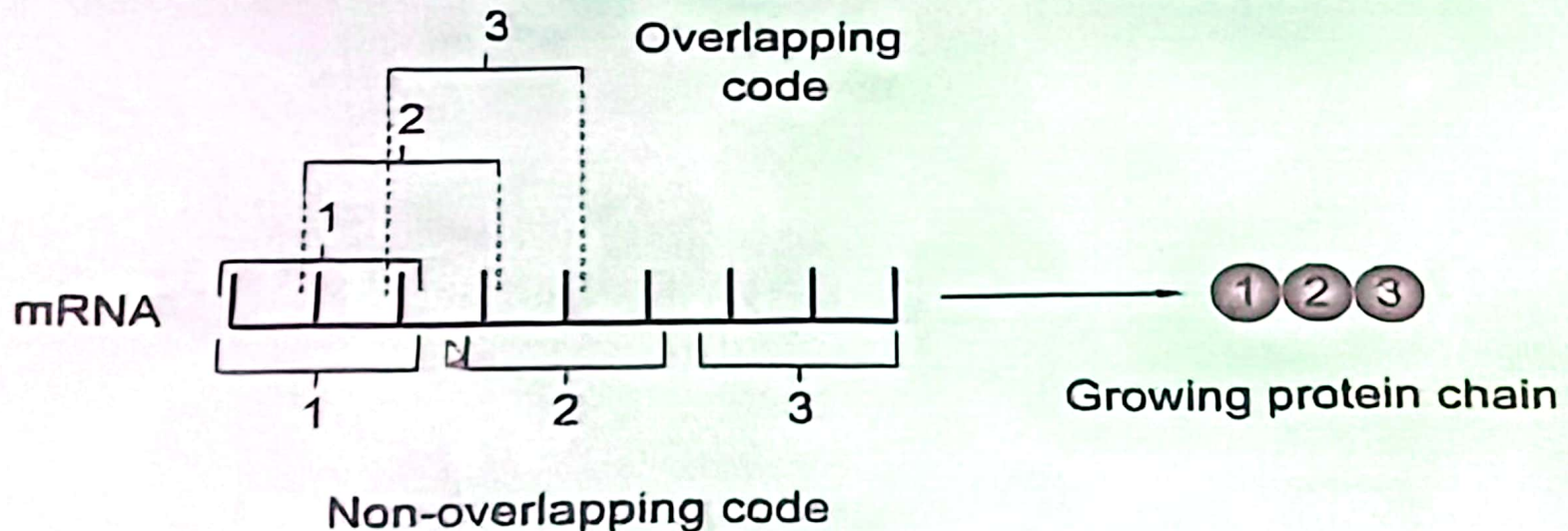
This is the language in which genetic information is stored.

Nature of Genetic code

1. The genetic code is triplet in nature.

Code words, or codons, each consisting of three ribonucleotides direct the insertion of a single amino acid into a polypeptide chain during its synthesis. AUG Methionin

2. The genetic code is ***nonoverlapping***, i.e. each triplet code is translated independently of its neighbours. AUG UGC GGG



Nature of Genetic code

3. The genetic code is ***comma free***, i.e. each codon in the mRNA is not separated from the neighboring codons by any nucleotides serving as commas or punctuation signals.

AUGUGCGGG

4. There are codons which code for ***initiation*** of the polypeptide chain as well as for its ***termination***.

AUG The initial amino acid inserted into all polypeptide chains in bacteria is ***N-formyl methionine (fmet)***, while it is ***methionine*** in eukaryotes..

Three other triplets, **UAA, UAG and UGA** serve as ***termination***



Nature of Genetic code

5. The genetic code is ***unambiguous***, i.e. any triplet cannot specify except one amino acid. AUG Methionin

GGG Glycine

6. The genetic code is ***degenerate***, i.e. there is more than one codon assigned for each amino acid.

Glycine GGG
GGC
GGA
GGU

7. The genetic code is ***universal***, i.e. the code applies equally to viruses, bacteria and eukaryotes.



Universality of the Genetic Code:

Applying equally to viruses, bacteria and eukaryotes.

The nature of mRNA and the translation machinery seemed to be very similar in these organisms.

For example, cell-free system derived from bacteria could translate eukaryotic mRNAs.

Many recent studies involving recombinant DNA technology have revealed that eukaryotic genes can be inserted into bacteria and transcribed and translated.



Certain exceptions, however, to the coding dictionary were discovered in the case of transcribed *mitochondrial DNA*.

This type of DNA, however, is extrachromosomal, and is related to *extrachromosomal inheritance*.



Genetic code

Brenner (1957) proposed the triplet code,

triplet code represent the minimal use of four nucleotides to specify 20 amino acids.

If the code was double, the four nucleotides, taken two at a time at all possible permutations would provide only

$4^2 = 16$ unique code words.

Four-letter code would yield

$4^4 = 256$ code words, which are much more than the code words needed for 20 amino acids.

The triplet code, however, provides $4^3 = 64$ possibilities, which can satisfy the



Code dictionary

61 triplet codon: amino acid assignments

Seond letter

		Seond letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } Stop UAG } Stop	UGU } Cys UGC } UGA } Stop UGG } Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gin CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG } Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Third letter

Degeneracy and Wobble

Degeneracy shows a certain pattern.

In a set of codes specifying the same amino acids, the first two letters are the same, with only the third differing.

Wobble hypothesis during translation, the first two positions of each codon must be very precise.

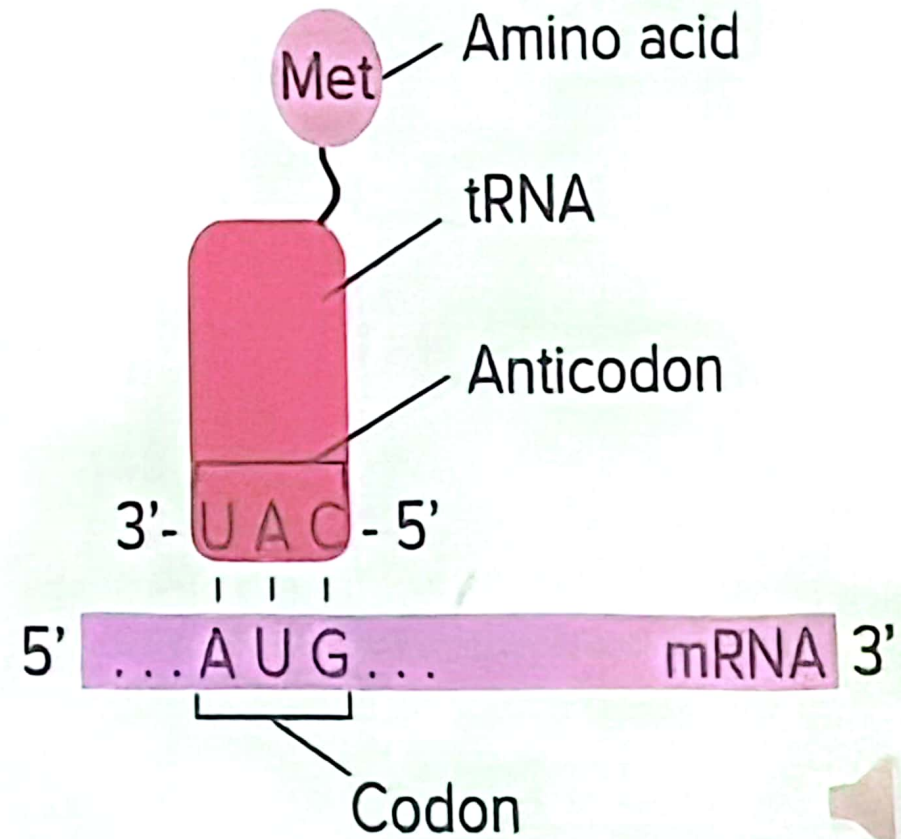
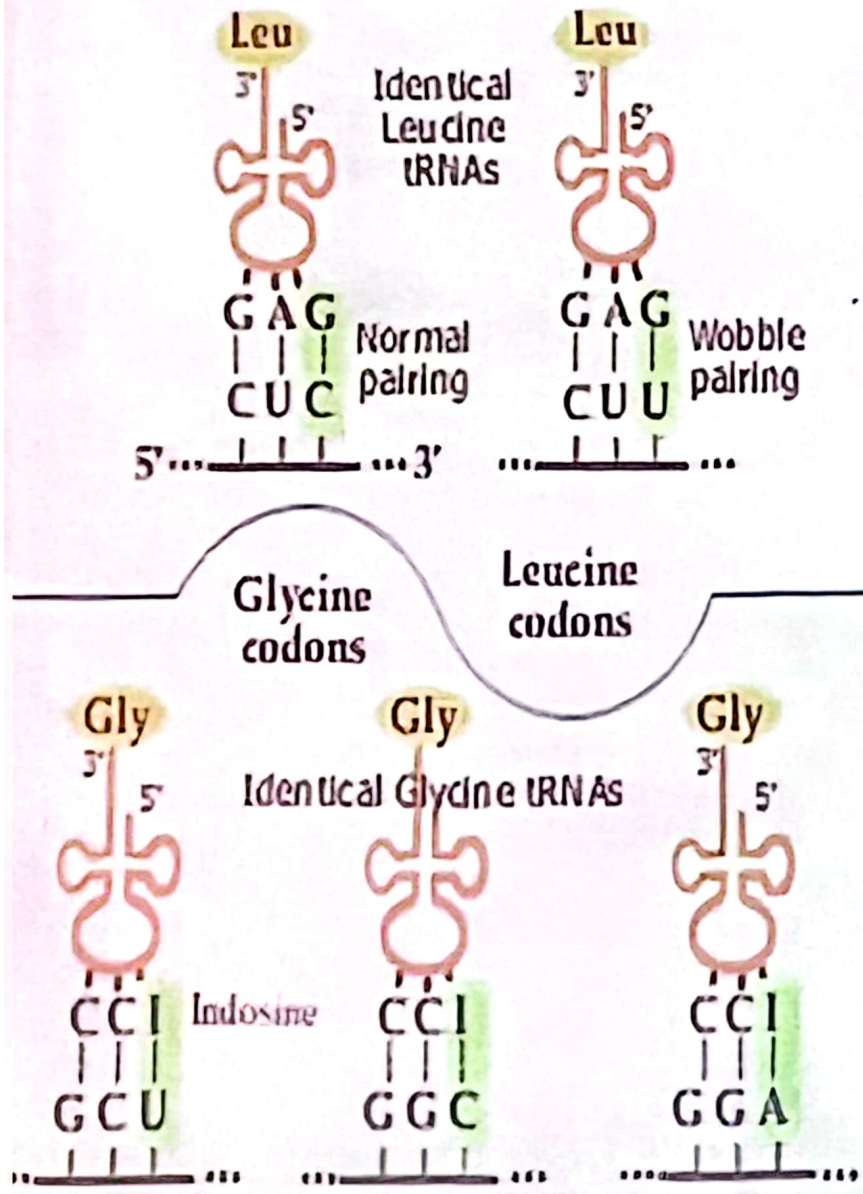
The third position, however, is often less specific. The wobble would allow the anticodon of a single tRNA type to pair with

	GGG
Glycine	GGC
	GGA
	GGU



Table 6.1. Base pairing according to wobble hypothesis.

First base of anticodon 5' end	Third base of codon 3' end
C	G
A	U
U	A or G
G	UC or A
I	UC or A



Glycine

GGG
GGC
GGA
GGU

↓
First position
of the tRNA

U
G
Inosine

↓
Third position
of mRNA triplet.

A or G
U or C.
C, U, or A.

Thus, A tRNA molecule of specific anticodon sequence can bind to more than one codon



Genetic expression

Genetic expression determines the structure and function of living organisms.

Its purpose is to cause specific reactions within the cell that leads to reactions between the organism and its environment.

These produce specific anatomical and physiological characteristics that distinguish this organism from other living organisms, in accordance with the genetic information stored in the genetic material received from its parents .



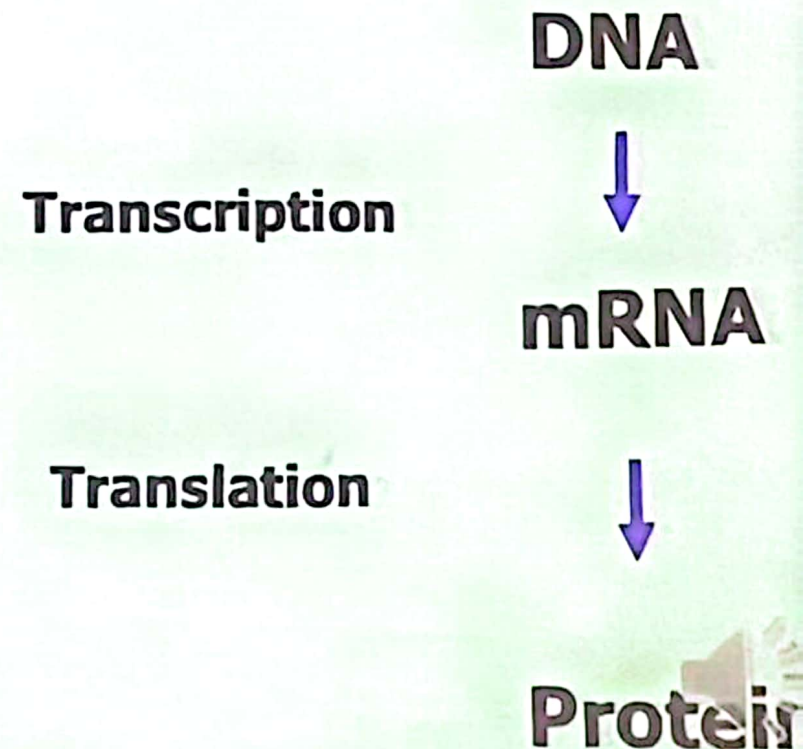
Genetic expression

- During the initial stage of gene expression, genetic information, stored in DNA, is transferred to RNA.

Protein synthesis is dependent on a series of
transfer RNA (tRNA)

Ribosome

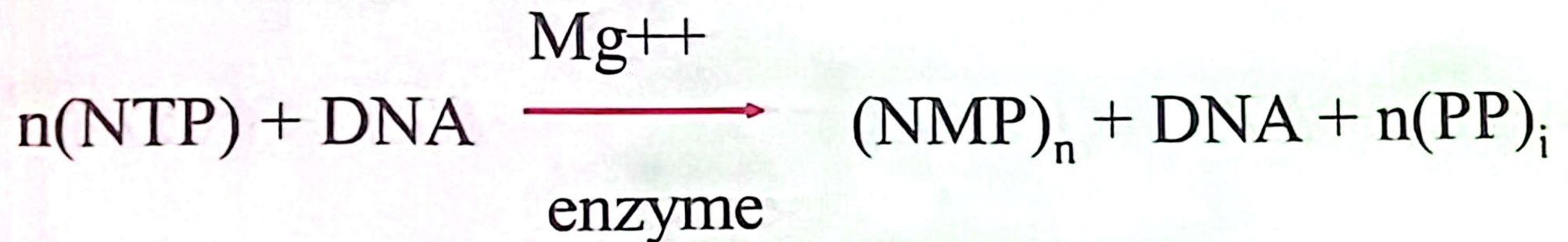
The process of transcription
and translation depend on
Base pairing affinity



RNA polymerase

This enzyme has the same requirements as DNA polymerase except that ribonucleoside triphosphates instead of deoxy ribonucleotide triphosphate

Its requirements



$n(\text{NTP})$ a certain number of ribonucleoside triphosphates

RNA polymerase

DNA = DNA template.
enzyme = RNA polymerase.

(NMP) = a sequence with the same number
of ribonucleoside monophosphates

$n(pp_i)$ = number of diphosphate groups released.



RNA polymerase Holoenzyme

(α , β , β' and σ). **sigma component**

Core
enzyme

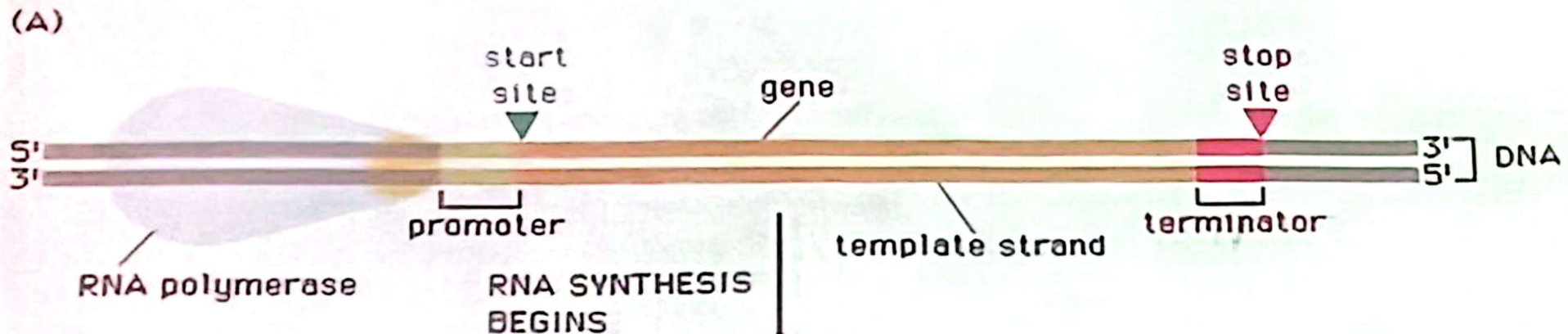
- Can be removed from the complex without loss of catalytic activity to the remaining *core enzyme*.
- play a regulatory function
- Involved in recognition of the points along the DNA template where RNA transcription is initiated, called *promoters*.



RNA polymerase

In eukaryotes, three different types of RNA polymerase are involved in the transcription of the three types of RNA: messenger, ribosomal and transfer.

In Prokaryotes, there is one enzyme for synthesis of the three types of RNA



Synthesis of RNA

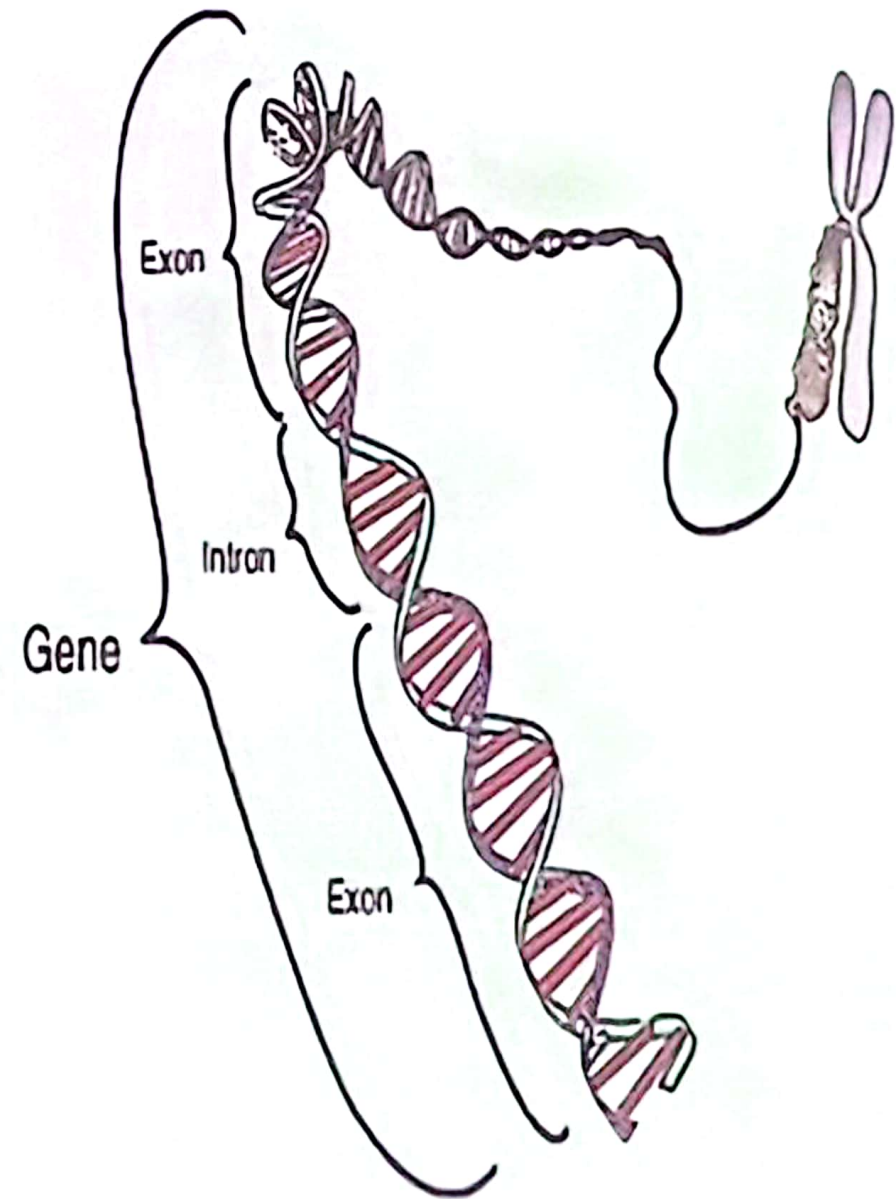
The process of transcription occurs on a DNA template that constitutes a gene.

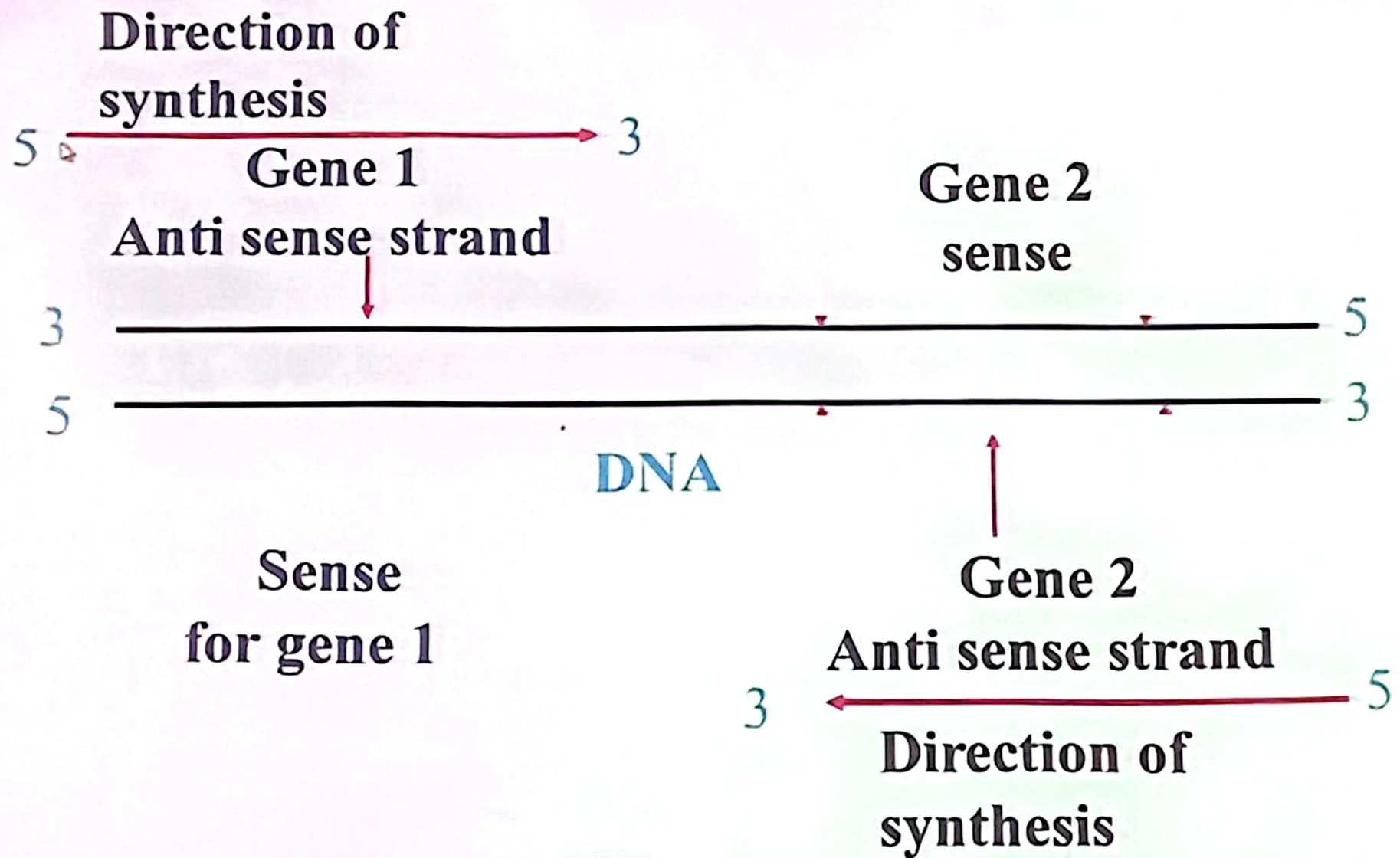
A Structural gene is defined as *a sequence of nucleotides which specifies the amino acid sequence (i.e. the structure) of a polypeptide chain*, and its transcript is called, after processing, a messenger RNA



A Structural gene

Its transcript is called
mRNA





Transcription is always in the direction 3_-to-5_ on the template antisense strand.
The synthesis runs on the RNA molecule from 5_-to-3_.

Sense and anti sense strand of DNA

Synthesis of RNA

The nucleotide sequence in any gene which is transcribed (i.e. serves as a template for RNA synthesis) is present on only **a part of one of the two strands** of a DNA double helix.

This strand is called the *antisense strand* or *coding strand*.

The complementary strand is called *sense strand*.

A DNA strand may contain the **antisense strand** of one gene but may constitute the **sense strand** (at some other location along the chromosome) of **another gene**.

Thus, some sequences on a DNA strand may function as **antisense strands** for certain genes, while some other sequences may serve as **antisense strands** for other genes



Synthesis of RNA

**1. Template binding,
3. chain elongation**

**2. chain initiation,
4. chain termination**

Synthesis of RNA

1. Template binding,
2. chain initiation,
3. chain elongation
4. chain termination

1-Template binding

- RNA polymerase binds at a region which precedes the gene sequence called a **promoter**.
- The promoter is recognized by the **sigma subunit** of the enzyme RNA polymerase.
- The promoter is not transcribed, but it acts as a signal for starting transcription.



Synthesis of RNA

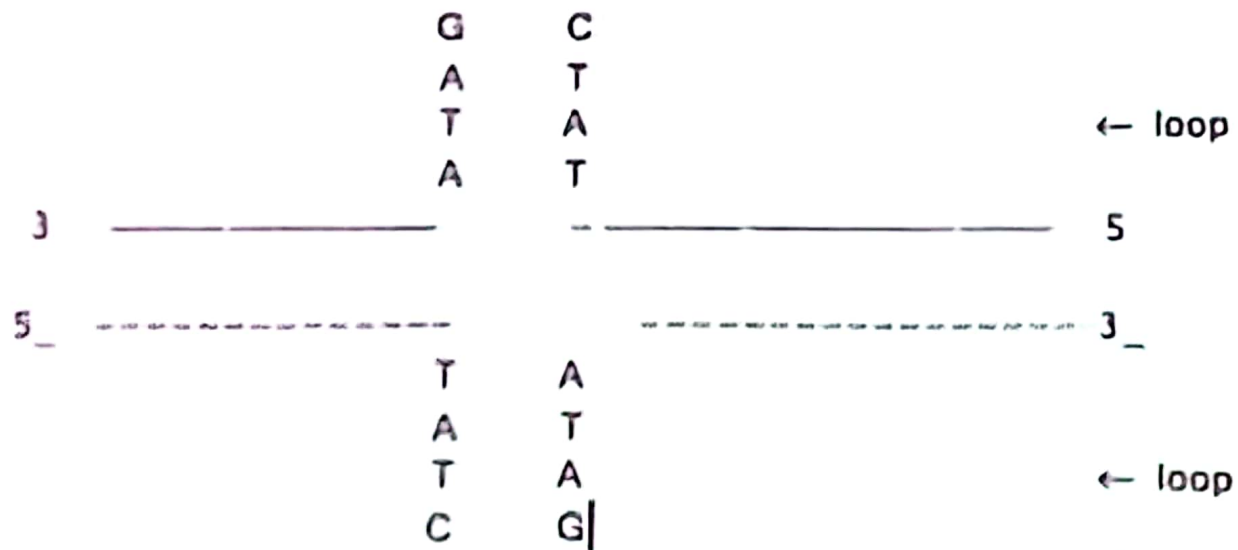
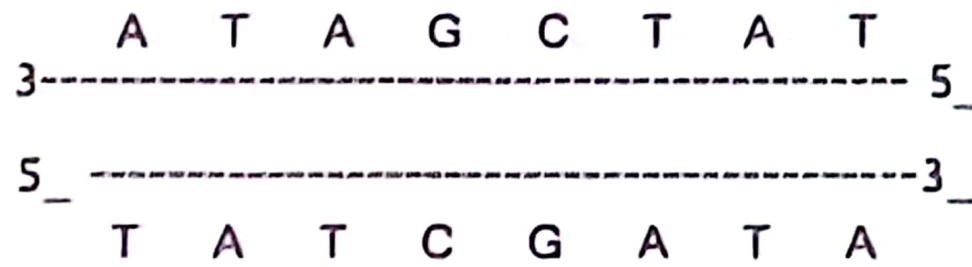
- Promoter regions of various viruses and bacteria have one part in common made up of a sequence of some nucleotides which is called Pribnow sequence.
- 3..A-T-A-G-C-T-A-T- ...5
- RNA polymerase binds strongly to the DNA template at Pribnow sequences. Then it moves in 3' to-5' direction for a distance of 10 nucleotides, after which RNA synthesis is initiated.



Synthesis of RNA

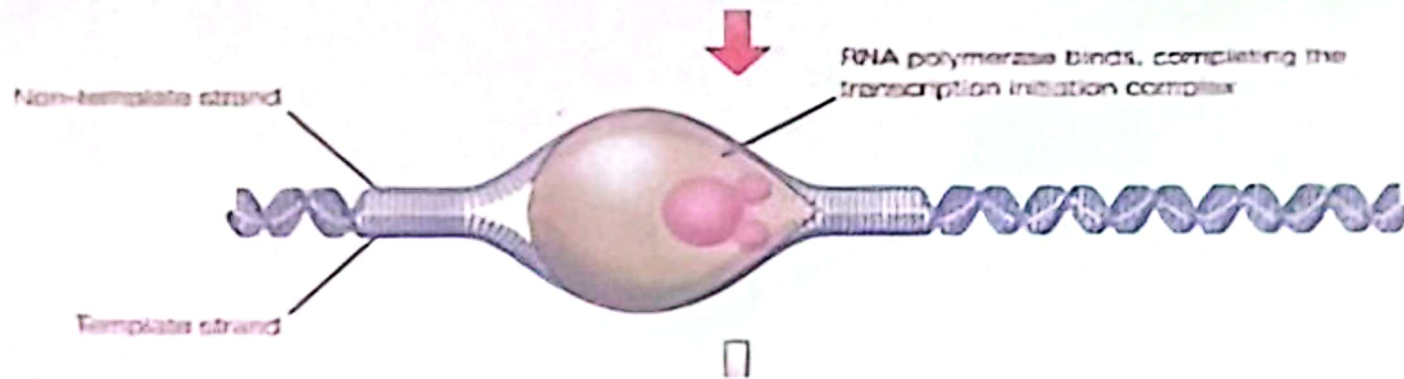
- It has been found that the two complementary strands at Pribnow sequences form inverted repeat sequences,
- Called **palindromic**.
- This allows the formation of a loop in the DNA duplex
- The formation of such a loop may facilitate the binding of the enzyme RNA polymerase to the antisense strand.
- Template binding leads to the formation of a complex structure made up of the enzyme and the promoter.



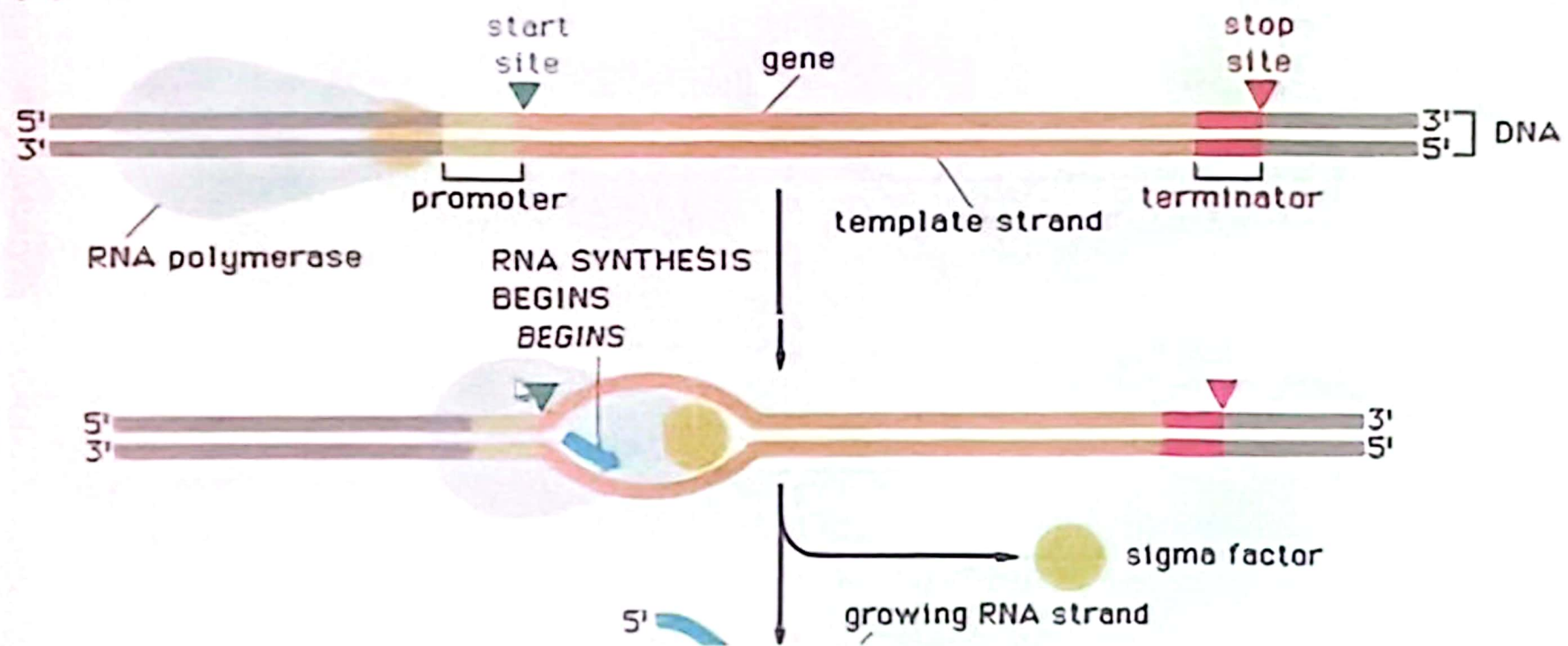


2. RNA Chain Initiation:

The RNA polymerase holoenzyme (**sigma subunit plus core enzyme**) moves along the DNA molecule until it meets the **first two nucleotides to be transcribed**.



(A)



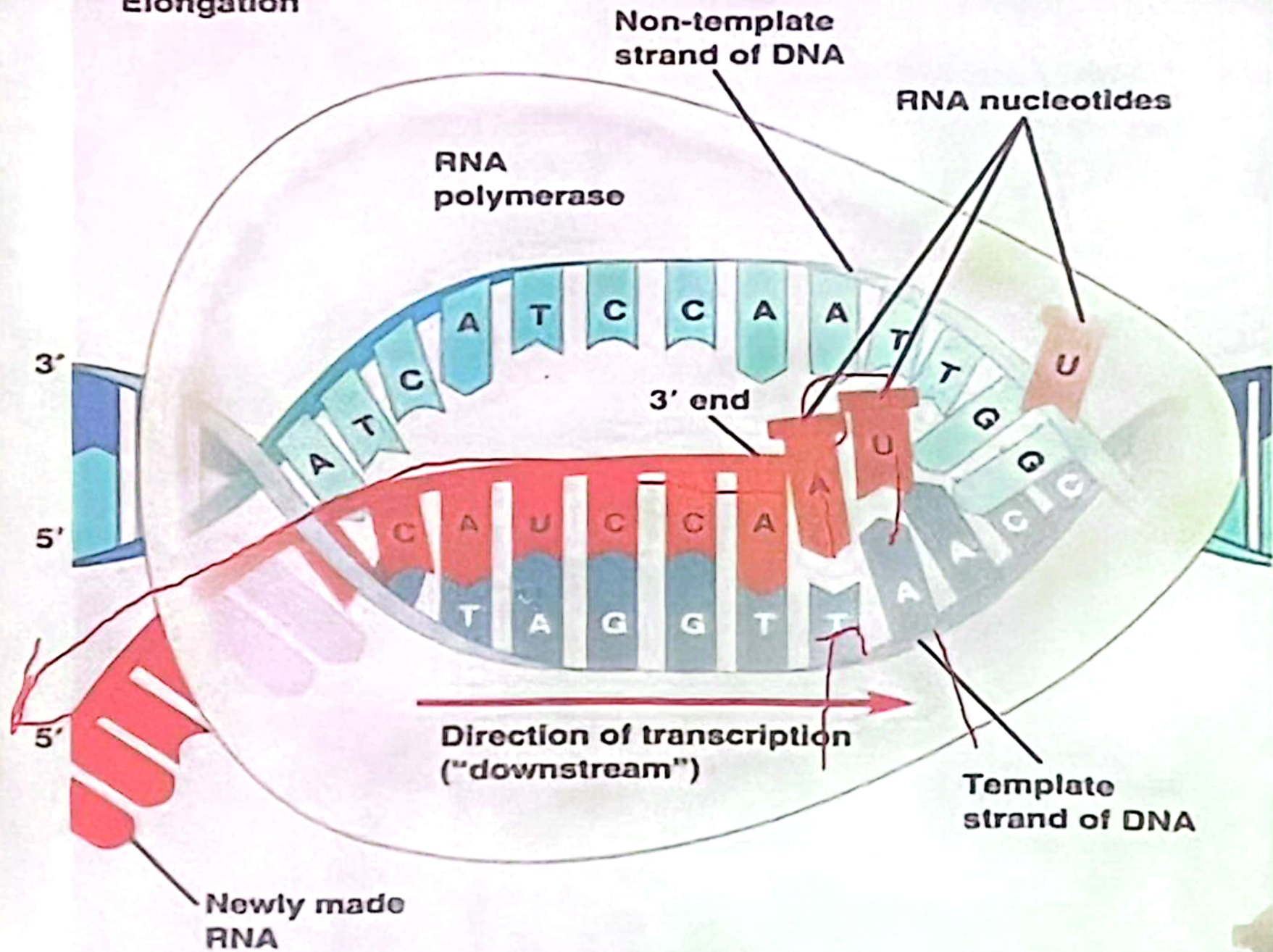
2. RNA Chain Initiation:

The **two complementary ribonucleotides** bind to the open-promoter complex at this site, and become linked together by a **phosphodiester bond**, forming a dinucleotide (a **chain of two ribonucleotides**).

This results in chain initiation.



Elongation



2. RNA Chain Initiation:

As the holoenzyme moves, the DNA helix ahead of it unwinds, while the helix behind it recloses.

Also, the newly synthesized RNA is released from its hydrogen bonds with the DNA as the helix reforms

3. RNA Chain Elongation:

As soon as the first two ribonucleotides have been linked, chain elongation proceeds rapidly in the 5'-to-3' direction, antiparallel to the DNA template antisense strand (i.e. along the 3'-to-5' direction of the antisense strand).



2. RNA Chain Elongation :

- After about 8 nucleotides are added to the growing RNA chain, RNA polymerase changes its structure and **losses the sigma subunit**, and elongation continues under the direction of the **core enzyme**.
- This process proceeds in *E. coli* at the rate of **50 nucleotides per second** at 37 oC.



2. RNA Chain Elongation :

- After about 8 nucleotides are added to the growing RNA chain, RNA polymerase changes its structure and **losses the sigma subunit**, and elongation continues under the direction of the **core enzyme**.
- This process proceeds in *E. coli* at the rate of **50 nucleotides per second at 37 oC**.
- The bonds between newly synthesized RNA and the antisense strand are dissociated, and RNA molecule grows in length until its termination.



2. RNA Chain Termination

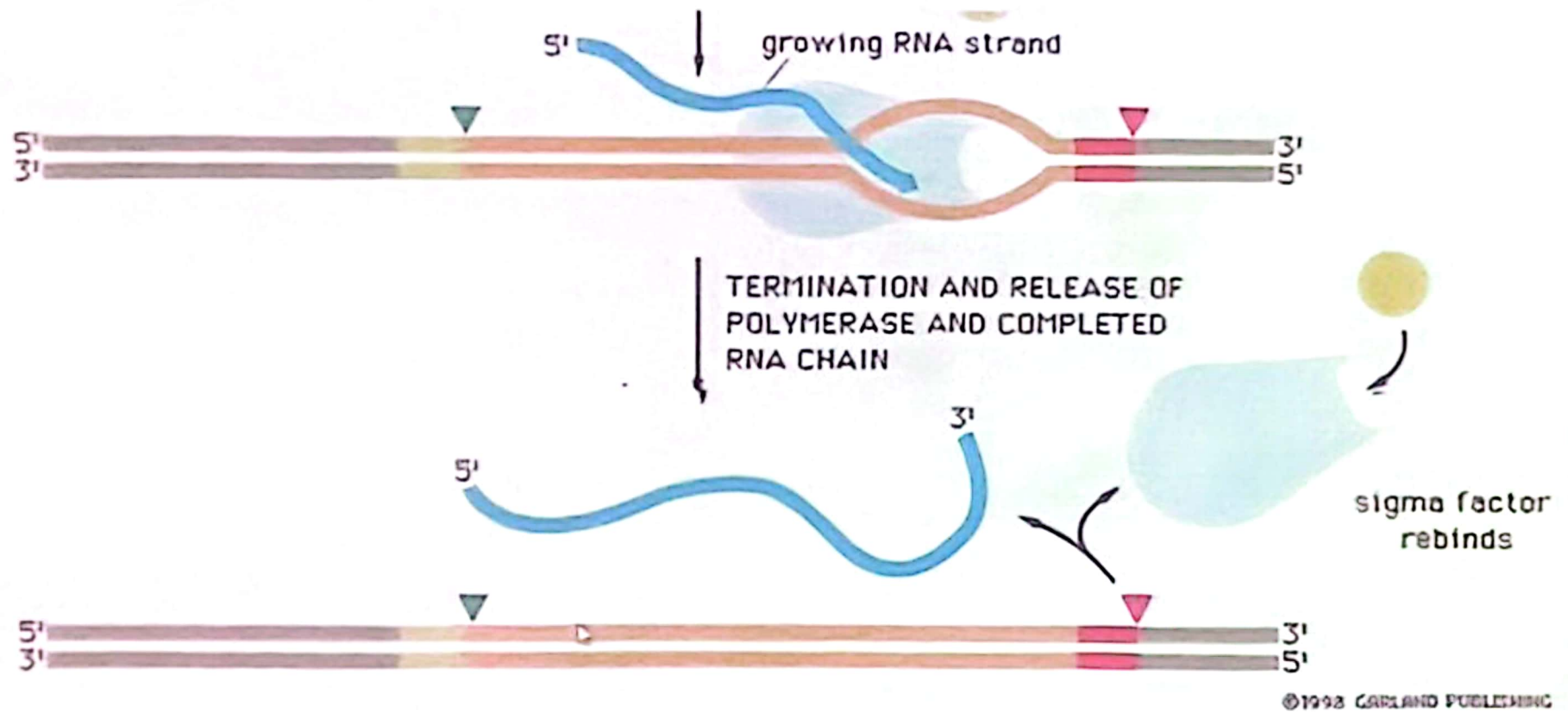
RNA synthesis stops when the core enzyme encounter a *termination signal* which consists of a specific nucleotide sequence in the template DNA, in the presence of a *termination factor* or rho.

The transcribed RNA molecule becomes completely released from the DNA template, and the core enzyme dissociates.

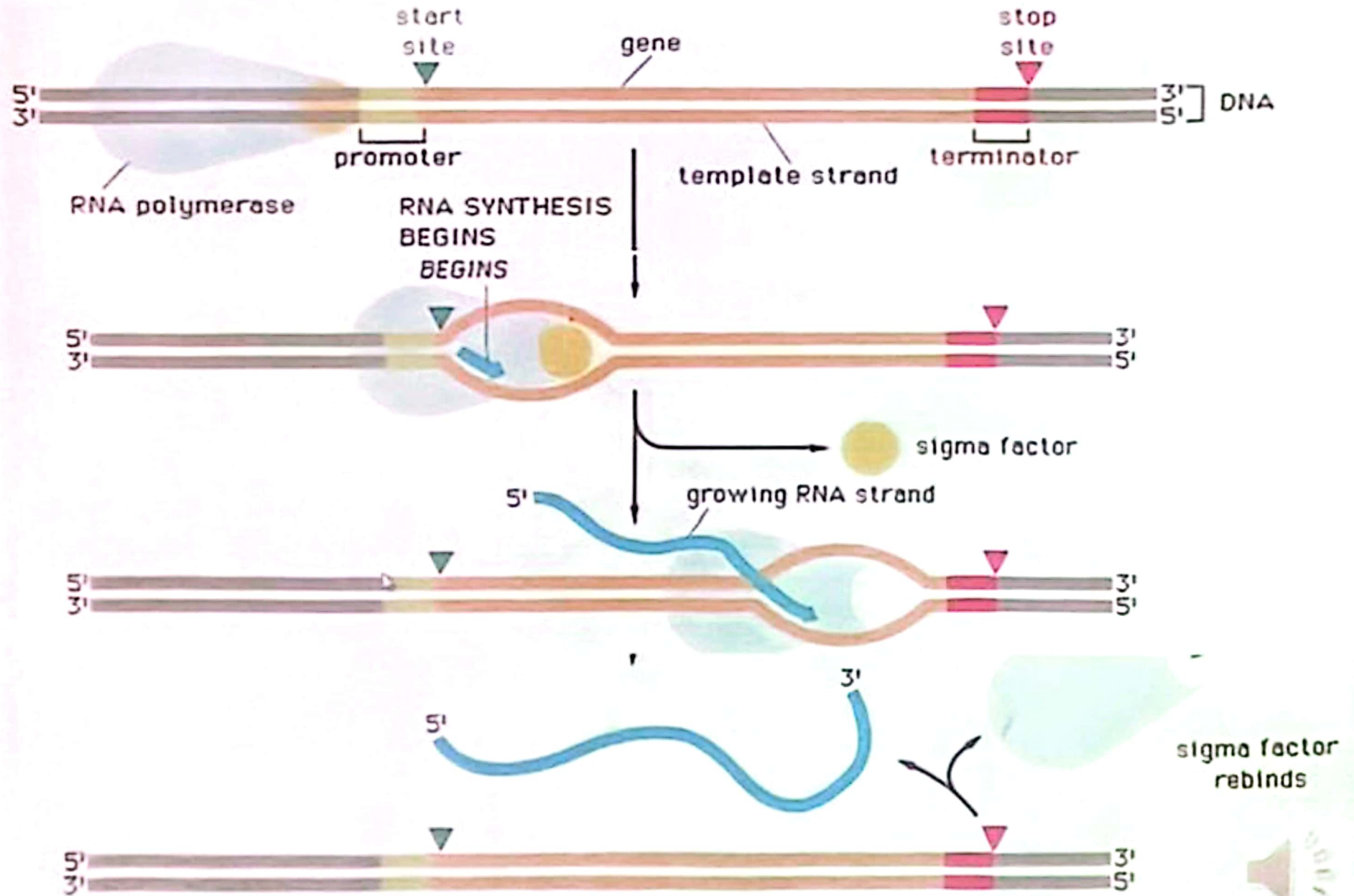
New RNA where:

A, T, C or G deoxyribonucleotide existed on the DNA
U, A, G, or C ribonucleotide has been added to the RNA molecule, respectively.





(A)



©1995 GARLAND PUBLISHING

Structure of RNA

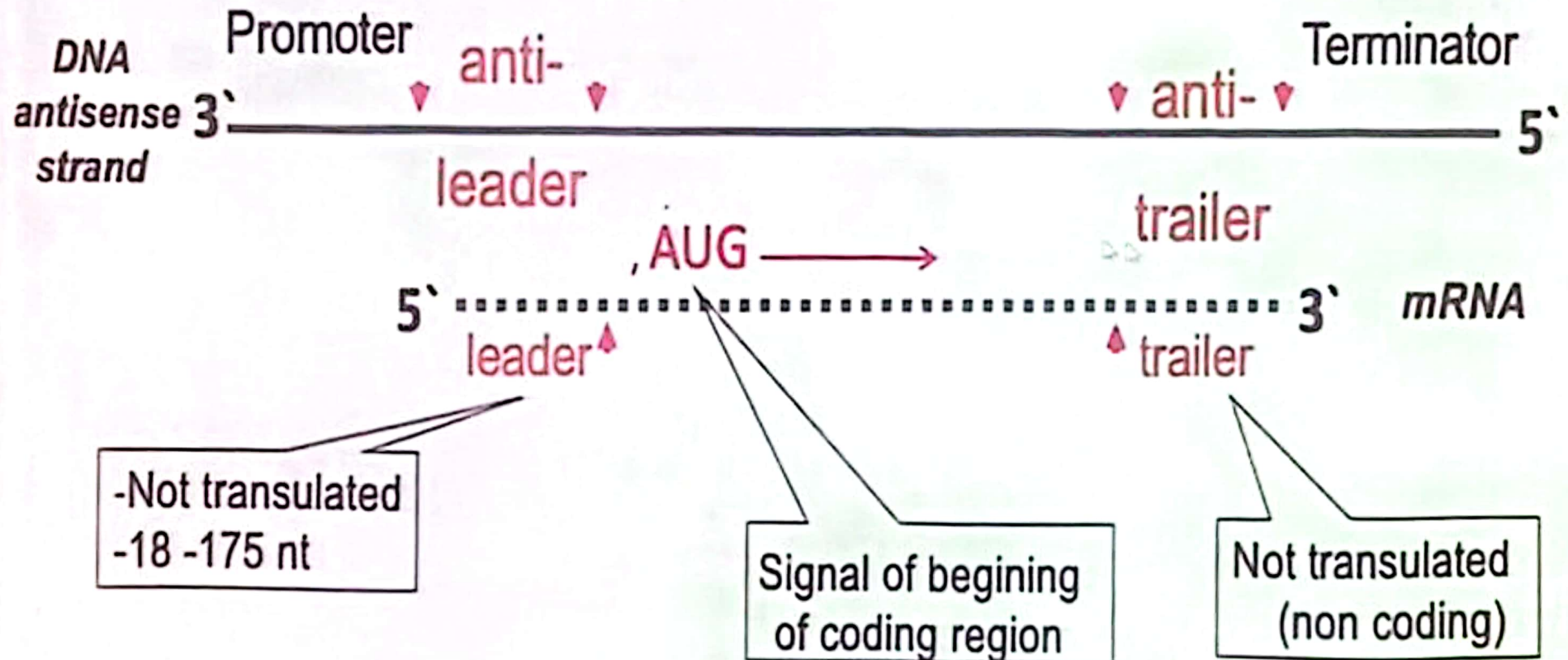
- An mRNA encoding a single polypeptide is called **monocistronic**.
- A cistron is a gene that specifies one polypeptide chain plus the start and stop signals.
- A **polycistronic** mRNA is an mRNA molecule that encodes several different polypeptide chains.
- In eukaryotes, all mRNA molecules are monocistronic.
- The size of an mRNA molecule depends on the length of the polypeptide or polypeptides it encodes, and thus may range from about 150 nucleotides to 12000 nucleotides.

Structure of RNA

- The newly synthesized RNA molecule starts with a **leader sequence** which **is not translated**, and which is **18 to 175 nucleotides** in length.
- Towards its end, the leader contains a specific sequence of **4 to 6 nucleotides** which are complementary to a sequence in the **ribosomal RNA** (16S RNA in E. coli).
- It is believed that this complementarity has a functional significance in the binding of the **ribosome to the mRNA** molecule before translation begins.

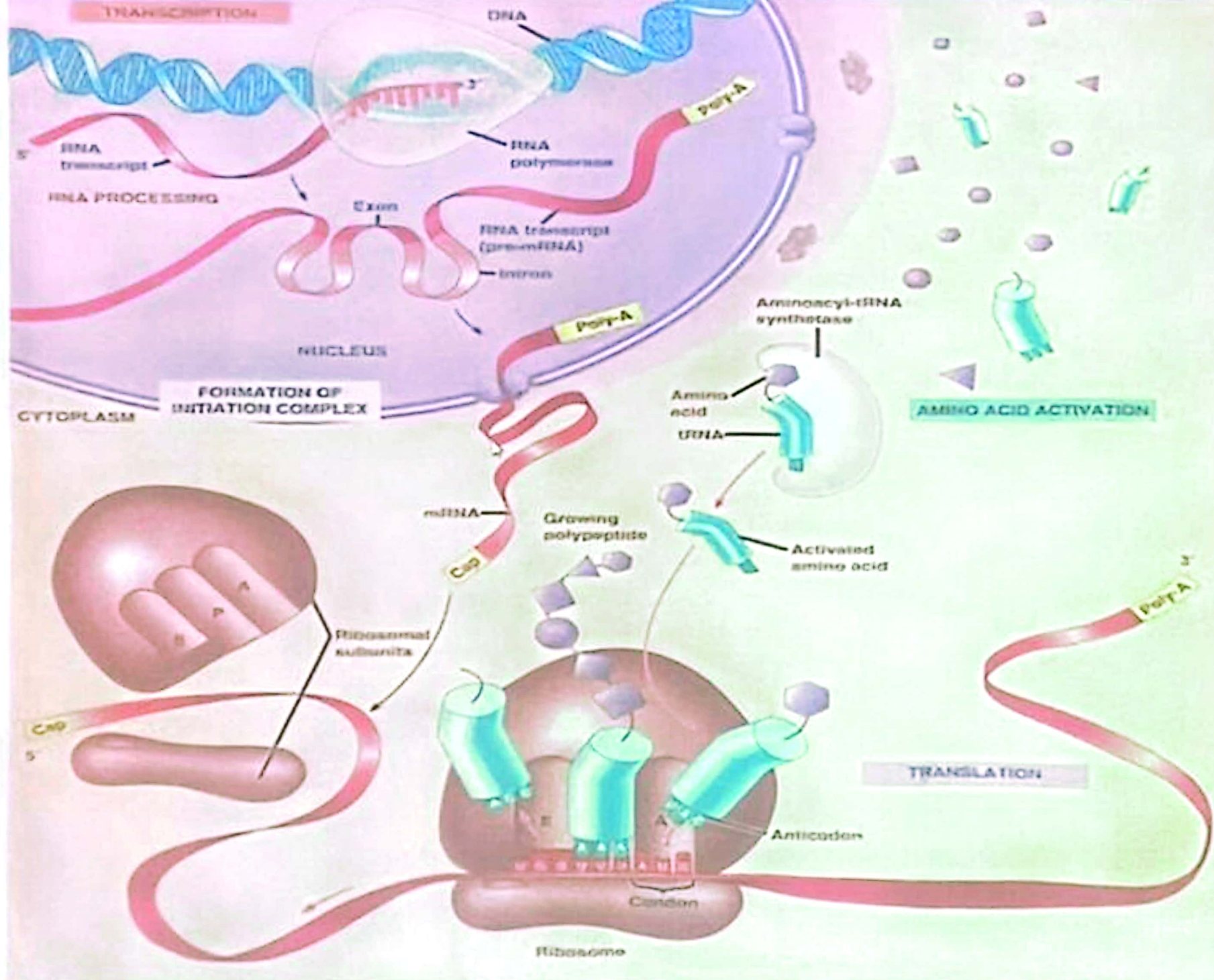


Structure of RNA



Structure of mRNA





Processing of mRNA

The newly synthesized RNA is subjected to certain modifications after transcription, called posttranscriptional modifications or processing.

Initial mRNA is called Pre-mRNA or heterogeneous RNA
hnRNA

Longer than cytoplasmic or processed mRNA

In bacteria, the relationship between DNA and RNA is direct. The DNA nucleotide sequence is transcribed into an mRNA sequences, which is translated into an amino acid sequence.

No processing of mRNA in prokaryots

In eukaryotes, complex processing of hnRNA occurs before the mature mRNA molecule appears in the cytoplasm.



Processing of mRNA

Processing include

1. Addition of Poly-A- tail

The initial RNA becomes polyadenylated, addition of 200 A-nucleotides (adenylic acid) tail at the 3'-OH, called poly-A tail.

Poly A tail protects mRNA from lysis in the cytoplasm

2. Addition of a cap:

At the 5' end of the RNA molecule (which has 3 phosphate groups) a *cap* is added, which is made up of *7-methylguanosine (7mG)*,

Cap facilitate outlet of mRNA from the nucleus to cytoplasm

5' Cap
7mGppp -Nmp -----A-A-A-A-A-A-3 poly-A tail



Processing of mRNA

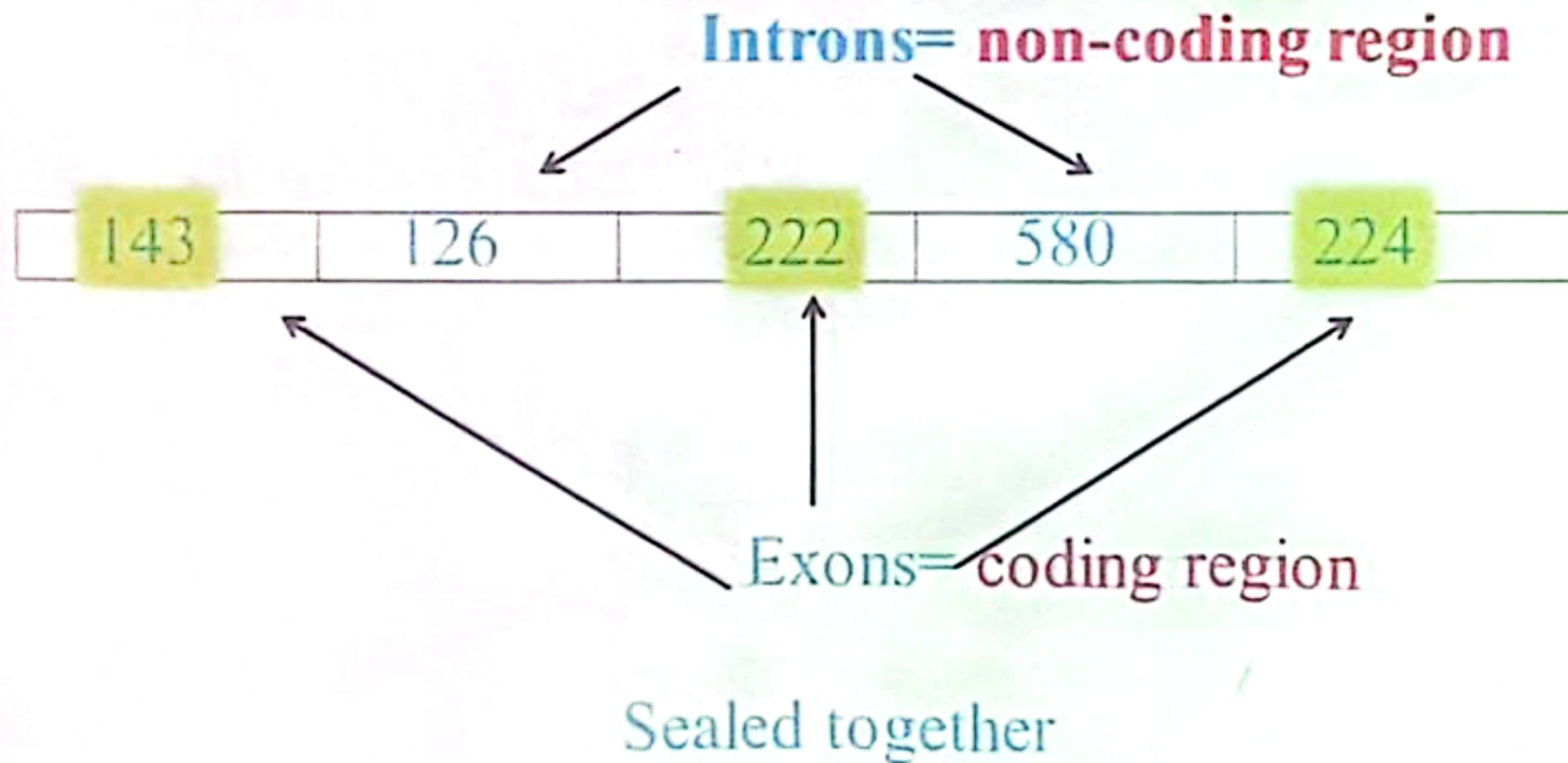
Excision of Intervening Sequences: Intervening sequences represent noncoding sequences called **introns**.

remaining coding sequences are called **exons**, which become splices (sealed) together to constitute the final mRNA.



Processing of mRNA

3. Excision of Intervening Sequences:



7mGppp -Nmp I

143		222		224
-----	--	-----	--	-----

 A-A-A-A--3 poly-A tail

